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Smart Plant Watering System

Project Overview

The main objective of this project is to design and implement an automatic plant watering system using a moisture sensor and a microcontroller. The system helps maintain the plant's health by automatically watering it whenever the soil becomes dry.

This project is useful for people who forget to water plants regularly and demonstrates the practical use of sensors, automation, and embedded systems.

Components Used

1. Arduino UNO
2. Soil Moisture Sensor
3. Water Pump
4. Relay Module
5. Breadboard
6. Jumper Wires
7. Water Tube
8. Plant Pot
9. Power Supply

Working Principle

The soil moisture sensor continuously measures the moisture level of the soil.

When the soil becomes dry and the moisture value drops below a certain threshold, the Arduino sends a signal to activate the water pump through the relay module.

The pump automatically supplies water to the plant until the soil moisture returns to a normal level. After that, the pump stops automatically.

This process allows the plant to water itself without human intervention.

System Flow

1. Read soil moisture value from the sensor
2. Compare the value with the predefined threshold
3. If the soil is dry
(Turn ON the water pump)
4. If the soil is wet enough
(Turn OFF the water pump)
5. Repeat the process continuously

Circuit Description

1. The soil moisture sensor is connected to the analog input pin of the Arduino.
2. The relay module is connected to a digital output pin.
3. The water pump is controlled through the relay.
4. The pump receives power from an external power source.

Results

The system was tested successfully using real soil and a small water pump.

When the soil became dry, the pump turned on automatically and watered the plant. Once the soil reached the required moisture level, the pump stopped.

The project successfully achieved automatic irrigation with minimal human effort.

Advantages

1. Saves time and effort
2. Prevents plants from drying out
3. Reduces water waste
4. Low-cost and simple design
5. Easy to expand for multiple plants

Challenges Faced

1. Adjusting the moisture threshold correctly
2. Stabilizing sensor readings
3. Managing water flow from the pump

Future Improvements

1. Add Wi-Fi monitoring using IoT
2. Display moisture percentage on an LCD screen
3. Use solar power for energy saving
4. Add a mobile application for notifications

Conclusion

This project demonstrates a simple and effective smart irrigation system using Arduino and a soil moisture sensor. The automatic watering process improves plant care while reducing manual work. The project also provides practical experience in sensors, automation, and embedded system applications.